

Homework 2, due Sep 3 in Brightspace before class (9am)

1. Spherical, static charge distribution has charge density

$$\begin{aligned}\rho &= (1 - r/r_0)\rho_0, \quad r \leq r_0 \\ \rho &= 0, \quad r \geq r_0\end{aligned}\tag{1}$$

Find electric potential. (Normalize potential so that $\phi(r_0) = 0$.) By differentiating potential find electric field.

Find total charge, $q(r)$ (charge within r), find electric field using Gauss theorem, compare.

Find total electromagnetic energy $\mathcal{E} = \int dV \frac{E^2}{8\pi}$. Also, verify $\mathcal{E} = (1/2) \int dV \rho\phi$.

Hint: do not forget about the surface term.

$$\begin{aligned}Q &= 4\pi \int_0^{r_0} \rho r^2 dr = \frac{1}{3}\pi\rho_0 r_0^3 \\ q(r) &= \frac{\pi r^3 (4r_0 - 3r)}{3r_0} \rho_0 \\ \Delta\phi &= \frac{1}{r^2} \partial_r (r^2 \partial_r \phi) = -4\pi\rho \\ \text{with } \phi(r_0) &= 0 \\ \phi &= \frac{\pi\rho_0 (r^3 - 2r_0 r^2 + r_0^3)}{3r_0} \\ \phi(0) &= \frac{1}{3}\pi\rho_0 r_0^2 \\ E_r &= \frac{\pi (4rr_0 - 3r^2)}{3r_0} \rho_0 \equiv \frac{q(r)}{r^2}, \quad r < r_0 \\ E_r &= \frac{Q}{r^2}, \quad r > r_0 \\ E_r(r_0) &= \frac{1}{3}\pi\rho_0 r_0 = \frac{Q}{r_0^2} \\ \mathcal{E}_{r < r_0} &= 4\pi \int_0^{r_0} \frac{E^2}{8\pi} r^2 dr = \frac{17}{630} \pi^2 \rho_0^2 r_0^5 \\ \mathcal{E}_{r > r_0} &= 4\pi \int_{r_0}^{\infty} \frac{E^2}{8\pi} r^2 dr = \frac{1}{18} \pi^2 \rho_0^2 r_0^5 \\ \mathcal{E}_{tot} &= \frac{26}{315} \pi^2 \rho_0^2 r_0^5 \\ \mathcal{E} &= (1/2) \int_0^{r_0} \rho\phi dV = \frac{17}{630} \pi^2 \rho_0^2 r_0^5 \neq \mathcal{E}_{tot}\end{aligned}\tag{2}$$

Problem? Potential at infinity is not 0. Surface term

$$\frac{4\pi}{8\pi} Q\Phi_{\infty} = \frac{1}{18} \pi^2 \rho_0^2 r_0^5\tag{3}$$

If we normalize potential to zero at infinity, or $\phi(r_0) = Q/r_0$, then

$$\begin{aligned}\phi &= \frac{\pi (r^3 - 2r_0 r^2 + 2r_0^3)}{3r_0} \rho_0, \quad r < r_0 \\ \mathcal{E} &= (1/2) \int_0^{r_0} \rho \phi = \frac{26}{315} \pi^2 \rho_0^2 r_0^5\end{aligned}\quad (4)$$

OK!

2. Consider a charge q_0 surrounded by a cloud with charge density

$$\rho_e = -\frac{q_0 e^{-\frac{2r}{a}}}{\pi a^3} \quad (5)$$

Find total charge of the system. Total potential $\phi(r)$, total electric field $\mathbf{E}(r)$. Plot $\phi(r)$, compare with Coulomb.

Hint:

$$\int e^{-x/a} x^2 dx = -a e^{-\frac{x}{a}} (2a^2 + 2ax + x^2) \quad (6)$$

For the point charge

$$\phi_0 = \frac{q_0}{r} \quad (7)$$

For the distributed charge

$$q = 4\pi \int_0^\infty \rho r^2 dr = -q_0 \quad (8)$$

$$\Delta \phi = \frac{1}{r^2} \partial_r (r^2 \partial_r \phi) = -4\pi \rho$$

$$\phi'(0) = 0$$

$$\phi(r = \infty) = 0$$

$$\frac{2\phi'(r)}{r} + \phi''(r) = \frac{4q_0 e^{-\frac{2r}{a}}}{a^3}$$

$$\phi = q_0 \left(e^{-\frac{2r}{a}} \left(\frac{1}{a} + \frac{1}{r} \right) - \frac{1}{r} \right)$$

$$\phi_{tot} = q_0 e^{-\frac{2r}{a}} \left(\frac{1}{a} + \frac{1}{r} \right)$$

$$E_r = \frac{q_0 e^{-\frac{2r}{a}} (a^2 + 2ar + 2r^2)}{a^2 r^2} \quad (9)$$

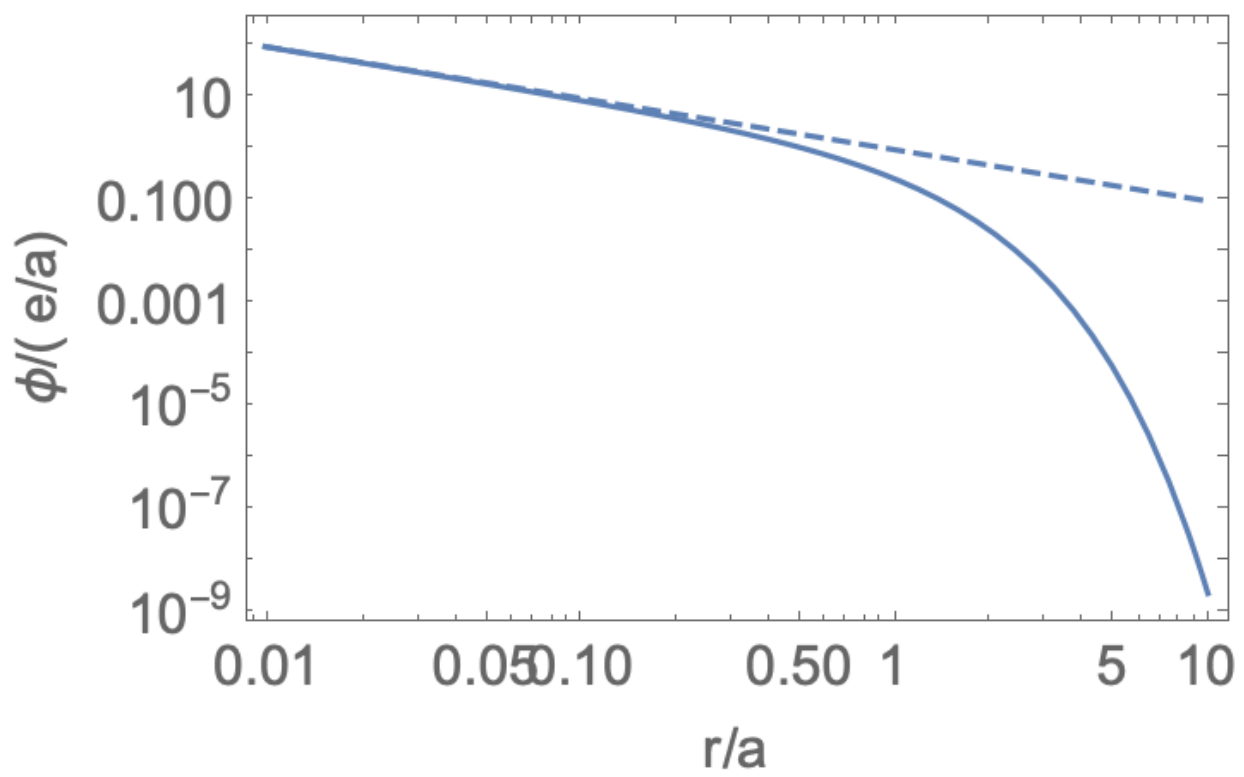


Fig. 1.— Electric potential of hydrogen atom; Dashed is Coulomb potential